Lijing LIU

Email: lliubo@connect.ust.hk **Phone:** 852 6093 7634/ 86 15392910588

EDUCATION

The Hong Kong University of Science and Technology | *Ph.D. in Mechanical Engineering* 09/2022 –Present

Supervisor: Prof. Rhea P. Liem, Imperial College London, Department of Aeronautics

The Hong Kong University of Science and Technology | *M.Sc. in Mechanical Engineering* 09/2021 – 06/2022

Supervisor: Prof. Zhigang Li, Hong Kong University of Science and Technology, MAE Department

Wuhan University of Technology | *B.Eng. in Mechanical Engineering* 09/2015 – 06/2019

RESEARCH INTEREST

- Aircraft design, multidisciplinary design and optimization, mission analysis (EOM-based simulation).
- Machine Learning (ML/DL/RL), data-enhanced modeling, surrogate modeling & application.
- Operations research, Scheduling and optimization of system resources, data analysis

PROJECT EXPERIENCE

Operation Driven Fleet Sizing and Allocation Optimization: Dynamic Strategy for Short-Term Aircraft Leasing

Funded by Hong Kong Research Grants Council (Project No. 16206022) – Research Part 1 09/2022 – Present

- **Institution:** Department of Mechanical and Aerospace Engineering, Hong Kong University of Science and Technology; The Brahmal Institute for Sustainable Aviation, Imperial College London.
- **Focus:** Development of a hierarchical multi-agent PPO reinforcement learning algorithm, fleet size planning and optimal task allocation methods within specified time intervals.
- **Outcomes:** Proposed methods for fleet planning and task allocation (publication 2), achieving deep coupling between aircraft design and airline operations (publication 1), enhancing practicality and operational efficiency (http://cerg1.ugc.edu.hk); multiple outcomes from this project are under preparation.

Operation-aware Aircraft Conceptual Design Optimization Framework for Sustainable Aviation

Funded by Hong Kong Research Grants Council (Project No. 16206022) - Research Part 2 09/2022 – Present

- **Institution:** Department of Mechanical and Aerospace Engineering, Hong Kong University of Science and Technology; The Brahmal Institute for Sustainable Aviation, Imperial College London.
- **Focus:** Developed a data-driven optimization framework for operation-aware aircraft concept design process to enhance the sustainability of airlines operation.
- **Outcomes:** Proposed an aircraft design optimization framework (publication 5) and continued to explore optimization for efficiency of the development-side models; multiple outcomes from this project are under preparation.

Time-Sequence Combination Optimization-Based Adaptive Computational Resource Scheduling 10/2024 – Present

- **Institution:** Department of Mechanical and Aerospace Engineering, Hong Kong University of Science and Technology & Huawei 2012 Laboratory - Hong Kong Research Center
- **Focus:** Conducting directed acyclic dependency decomposition of application components based on temporal information in multi-core computing scenarios; optimizing multi-core CPU resource scheduling for specified computational supply conditions.
- **Outcomes:** Gained a deep understanding of computer process/thread dependency structures, decomposition and acceleration of the application startup process.

Research on machine learning-based major diseases detection 12/2021 – 03/2022

- **Supervisor:** Prof. Shlomo Ta'asan, Carnegie Mellon University
- **Focus:** Built medical detection models using machine learning algorithms (decision trees, KNN, random forest, naive Bayes) on Kaggle disease datasets; Conducted systematic literature review of existing research
- **Outcomes:** Submitted papers to ML/AI conferences (Publication 6)

Deep Neural Network-Based Object Recognition Algorithm for Robotic Scenes 09/2019 – 08/2020

- **Institution:** School of Mechanical Engineering, Wuhan University of Technology
- **Focus:** Proposed innovative algorithm combining neural networks with skip-connection pyramid model to address small object recognition challenges; Pre-trained model on ILSVRCCLS-LOC dataset and validated on PASCAL VOC and MSCOCO datasets; Developed prototype object detection system using C++ MFC framework
- **Outcomes:** Published a journal paper (Publication 5), gained expertise in neural networks and C++ (MFC)

PUBLICATIONS & PATENT

Journal Articles & Conference Papers

- **Liu, L.**, Wu Yiran, M. A. Hariansyah, Rhea P. Liem*. "Bilevel Strategy for On-Design eVTOL Sizing and Off-Design Structural Weight Optimization" Manuscript for *ICAS2026* **(Accepted)**.
- **Liu, L.**, Reyner, C., and Liem, R. P. "Co-design of Aircraft Configuration and Flight Profile for Match-range Operation." *Aeronautics and Astronautics* (2025). (**Accepted**)
- **Liu, L.**, Reyner, C., and Liem, R. P. "Operation-based Fight Profile Optimization with Continuous Cruise Climb Operations." *AIAA Aviation 2025 Forum*. 2025.
- Reyner, C., **Liu, L.**, and Liem, R. P. " Integrating Wind-Aware Optimal Turning Maneuver Strategy into Operations for Fixed-Wing Unmanned Aircraft Vehicle (UAV)." *AIAA Aviation 2025 Forum*. 2025.
- **Liu, L.**, Kim, D., Reyner, C., and Liem, R. P., "Data-Driven Multi-Range Mission-Based Overall Aircraft Conceptual Design Optimization," *Proceedings of the 34th Congress of ICAS*, Florence, Italy, 2024.
- **Liu, L.** "A Deep Neural Network-Based Target Recognition Algorithm for Robot Scenes." *Scientific Programming* 2022 (2022): 1-14.
- **Liu, L.,** Gao Wentao "Research on several major diseases based on machine learning models", *2022 5th International Conference on Advanced Electronic Materials, Computers and Software Engineering (AEMCSE)*. IEEE, 2022.
- **Liu, L.,** "Discussion on the Integration of Mechatronics Engineering and Artificial Intelligence Technology." *Advances in Computer, Signals and Systems* 6.1 (2022): 29-33.

Manuscripts in preparation/process

- Qiyu Yan, Liu Lijing, Li Zhicheng. "AutoOR: An LLM-Enabled Automated Modeling Workflow for General Operations Research Problems" Manuscript for *IJCAI* **(submitted)**.
- **Liu Lijing,** James M shihua, Qiyu Yan, Rhea P. Liem*. "LLM-enhanced Dynamic Fleet Planning with Hierarchical Multi-agent Reinforcement Learning Framework" Manuscript for *IJCAI* **(submitted)**.
- **Liu Lijing,** Christian Reyner, Dajung Kim, Go Nam Lui, Rhea P. Liem*. "Operation-aware Aircraft Conceptual Design Optimization Framework for Sustainable Aviation" Manuscript under preparation.
- **Liu Lijing,** James M shihua, Christian Reyner, Rhea P. Liem*. "Operation-Driven Airline Fleet Sizing with Optimal Assignments: A Dynamic Strategy for Short-Term Aircraft Leasing" Manuscript under preparation.

Patent

- **Liu Lijing,** et al. (2025). Application Startup Method and Device. CN202510442458.0.

INTERNSHIP EXPERIENCES

Research Intern | Huawei 2012 Lab - Hong Kong Research Center, Hong Kong SAR 10/2024– 04/2025

- **Cold-Start Acceleration Module:** Led the design and implementation of a preloading module using application process preloading and IO file prefetching to reduce PC applications launch latency. Developed architecture, wrote production-level code, and debugged independently. Resulted in a patent (1st inventor).
- **Dynamic Control Optimization:** Built a user behavior prediction model to dynamically adjust preloading strategies, improving latency reduction by more than 20% under constrained memory.
- **Full-Stack Development:** Owned end-to-end delivery (design → testing) of both modules, enabling 2 internal project conversions and securing 1 patent + 1 company technical secret, which were recognized as a potential high-value project in July 2025.

Scientific Software Developer | iFlytek StarAI Tech, Wuhan 07/2020– 12/2020

- **Task Management System:** Designed system architecture and developed core features (task creation/assignment/tracking) using Python, Java, and JavaScript, ensuring seamless database integration.
- **Cross-Functional Collaboration:** Coordinated with testing teams to resolve 15+ critical bugs, delivering the system on time with 99.8% operational stability.

ADDITITIONAL INFORMATION

Technical Skills

- **Programming:** Python, C/C++, Java, JavaScript (jQuery, HTML), SQL
- **Tools:** AutoOR (in-house LLMs workflow for OR problem), ANSYS, IntelliJ IDEA, QuTiP, MFC
- **Frameworks:** TensorFlow (implied via Deep Learning projects), Langchain (Entry-level)